

General Pharmacology & Dental Therapeutics

General Pharmacology & Dental Therapeutics

Departmental Objectives:

- At the end of the course the student will be able to
- Describe the principles of rational prescribing and the basis of therapeutic decision making.
- State the principles underlying the concepts of essential medicines.
- Recognize the implications of polypharmacy and other means of irrational prescribing.
- Describe reactions, interactions and manage problems due to misuse and abuse of medicines.
- Demonstrate knowledge and understanding of teratogenic medicines.
- Select appropriate learning resources periodically.
- Evaluate the ethical and legal issues involved in drug prescribing
- Develop attitude for continuing self learning.
- Describe the pharmacokinetics and pharmacodynamics of essential and commonly used drugs in dentistry
- List the indications, contraindications, interactions and adverse reactions of commonly used drugs in dentistry with reasons
- Tailor the use of appropriate drugs in disease with consideration to its cost, efficacy, safety for individual and mass therapy needs
- Prescribe common and essential drugs in special medical situations such as pregnancy, lactation, old age, renal or hepatic damage and immuno-compromised patients

List of competencies to acquire:

- Therapeutic decision making.
- Writing rational prescription.
- Management of common dental infections.
- Management of dental pain.
- Proper use of local anesthetic agents.
- Prescribing from essential medicine list.
- Providing Chemoprophylaxis.
- Describe indications, contraindications and side effects of medicines related to dental practice.
- Management of Shock
- Prescribe drugs for common dental problems
- Demonstrate knowledge and understanding of adverse reactions and drug interactions of commonly used drugs in dentistry
- Critically evaluate drug formulations and interpret the clinical pharmacology of marketed preparations commonly used in dentistry

Distribution of teaching /learning hours

Subjects	Lecture	Tutorial	Practical + Clinical	Total Teaching hours	Integrated teaching (Common)	Formative Exam		Summative exam	
						Preparatory leave	Exam time	Preparatory leave	Exam time
General Pharmacology	50 hrs	35 hrs	22 hrs	210	10hrs	10 days	20 days	10 days	25 days
Dental Therapeutics	50 hrs	35 hrs	12+6 hrs						

Teaching/learning methods, Teaching aids and evaluation

Teaching Methods			Teaching aids	In course evaluation
Large group	Small group	Self learning		
Lecture Seminar	Tutorial Discussion Question answering session Practical	Assignment, Self study	Laptop, Computer & multimedia OHP, Transparency & Marker White board & Marker, Black board & chalks, Flip Chart, Slide projector Video, X-ray plate, View Box Model, Television, VCR, Cassette, Specimen, Analysis report	Item examination(oral) Card completion Examination (Written) Term examination Term final Examination (Written, oral & practical)

Related equipments:

Local Anaesthetic Cartidge syringe (Metallic)

1. Rubber Dam, Reamers, Files, Apex Locator
2. Rotatory Instruments used For Endodontics, including Endodontic Instruments
3. Condensor
4. Filling Materials, Amalgam Gun
5. Autoclave, Hot Air Woven
6. Glass Bead Sterilizer
7. Disposable Plastic Syringe
8. Bottles of Anesthetics
9. Anesthetic Spray For Topical application
10. Flouride Applicator
11. Tube For Topical Gel
12. Spirit Lamp
13. Stainless Steel Tray (Large, Medium & small)
14. Inhalation Agents & Container
15. Gingival Retraction Cord/ Liquid.

Professional Examination:

Marks distribution of Assessment of General & Dental Pharmacology

(Total marks 300)

- Written = 100 (SAQ=70 + MCQ =20 + Formative Assessment =10)
- SOE = 100
- Practical = 100

Learning Objectives and Course Contents of General Pharmacology

Learning Objectives	Contents	Teaching Hours
Students will be able to- ● Define Pharmacology, branches of pharmacology, Drug, doses, therapeutic index and mention the sources of drugs. ● State the routes of drug administration ● Describe absorption of drugs, processes and factors modifying drug absorption. ● Describe distribution of drugs, processes and factors modifying drug distribution ● Describe, aim and describe factors modifying biotransformation. ● State the processes and routes of drug elimination. ● State the mechanism of drug action and dose-response relationship. ● State the Drug interaction, Drug combination and Drug antagonism. ● State and identify adverse drug reaction ● Write prescription and mention legal, ethical and economic aspects. ● Identify the preparation of various formulations. ● Write prescriptions on common dental problem	1. Basic concept of pharmacology ● Pharmacology, branches of pharmacology, Drug, doses, therapeutic index and sources of drugs. ● Routes of drug administration, processes and factors modifying drug absorption, and processes and factors modifying drug distribution ● Factors modifying biotransformation. And processes and routes of drug elimination. ● Mechanism of drug action and dose-response relationship. Drug combination and Drug antagonism. adverse drug reaction ● Prescription writing and legal, ethical and economic aspects.	L-10 hours T-9 hours P-7 hour
Students will be able to ● Define neurotransmitter. Neurotransmission, mention criteria of neurotransmitter and classify autonomic receptor. ● Describe cholinergic drugs-classification, pharmacokinetic, pharmacodynamic, indication, contraindication and adverse drug reaction. ● Describe Anticholinergic drugs-classification, pharmacokinetic and pharmacodynamic, indication, contraindication and adverse drug reaction. ● Describe clinically important adrenergic drugs. ● List the clinically important alpha and beta blockers and their indication, contraindication and side effects.	2. Drugs acting on autonomic nervous system ● Neurotransmitter. Neurotransmission, criteria of neurotransmitter and autonomic receptor. ● Cholinergic drugs-classification, pharmacokinetic, pharmacodynamic, indication, contraindication and adverse drug reaction. And anticholinergic drugs-classification, pharmacokinetic and pharmacodynamic, indication, contraindication and adverse drug reaction. ● Clinically important adrenergic drugs. Clinically important alpha and beta blockers and their indication, contraindication and side effects.	L-7hours T-4 hours P-7 hours

Learning Objectives	Contents	Teaching Hours
Students will be able to- ● State the role of histamine in health and disease. ● Classify anti histamine, their pharmacological effects, indications, contraindications. and toxicity ● Describe eicosanoids-PGs	3. Autacoids ● of histamine in health and disease. ● Classification of anti histamine, their pharmacological effects, indications, contraindications. and toxicity ,and eicosanoids-PGs	L-1 hour
● Define and classify sedative and Hypnotic and its mechanism,indicationcontraindication and toxicity. ● Define and classify analgesic and their pharmacological effects,indication,contraindication and toxicity.[opioids and NSAIDs] ● Define and classify general and local anaesthetic and mention their mechanism of action,indication,contraindication and toxicity	4. Drugs acting on central nervous system ● Sedative and Hypnotic and its mechanism, indicationcontraindication and toxicity. ● Analgesic and their pharmacological effects, indication, contraindication and toxicity.[opioids and NSAIDs] ● General and local anaesthetic and mention their mechanism of action,indication,contraindication and toxicity.	L-8 hours T-10 hours
● Classify antihypertensive drugs. ● State the management of hypertension. ● List anticoagulants and antiplatelets and mention their mechanism, indication, contraindication and toxicity ● Mention commonly used diuretics. ● List the anti angina drugs[basic knowledge].	5. Drugs acting on cardiovascular and renal system ● Classification of antihypertensive drugs. ● Management of hypertension. ● anticoagulants and antiplatelets and their mechanism, indication, contraindication and toxicity name of commonly used diuretics and anti angina drugs[basic knowledge]	L-5 hours T-5 hours
● Mention the treatment of peptic ulcer. ● Mention the management of diarrhea and emesis	6. Drugs acting on Gastrointestinal system	L-3 hours

Learning Objectives	Contents	Teaching Hours
Students will be able to ● Classify anti-diabetic drugs and mention their mechanism, indications, contraindication and toxicity. ● List the steroids and mention their mechanism, kinetics, indication, contraindication and toxicity	7. Endocrine pharmacology ● Classification of anti-diabetic drugs and their mechanism, indications, contraindication and toxicity. ● Steroids and their mechanism, kinetics, indication, contraindication and toxicity.	L-2 hours
● List the anti-microbial drugs and mention their mechanism and resistance. ● Mention the general principles of use of anti-microbial drugs. ● Classify penicillin and cephalosporins and mention their indication, contraindication and toxicity. ● List tetracycline and mention their indication, contraindication and toxicity of tetracycline. ● List Macrolides and Quinolones and mention their indication, contraindication and toxicity. ● List the anti-tubercular drugs and mention toxicity and management of tuberculosis in short. ● List the antiamoebic drugs and mention the indication, contraindication and toxicity of metronidazole. ● List the anthelmintic and antifungal drugs and mention their toxicity and management of helminthiasis and fungal infection. ● State anti cancer drugs	8. Antimicrobial Agents ● Anti-microbial drugs and their mechanism and resistance. ● The general principles of use of anti-microbial drugs. ● Classification of penicillin and cephalosporins and their indication, contraindication and toxicity. ● Tetracycline and their indication, contraindication and toxicity. ● Macrolides and Quinolones and their indication, contraindication and toxicity. ● The anti-tubercular drugs and toxicity and management of tuberculosis in short. ● Antiamoebic drugs and indication, contraindication and toxicity of metronidazole. ● Anthelmintic and antifungal drugs and their toxicity and management of helminthiasis and fungal infection. ● anti cancer drugs.	L-12 hours T-9 hours
● State essential drug and mention the criteria of rational use of drugs. ● Mention the guide line of a rational prescription. ● Selection of P-drug.	9. Essential drug concept and rational drug use	L-2 hours P-6 hour

Learning Objectives and Course Contents in Dental Therapeutics

Learning Objectives	Contents	Teaching Hours
Student will be able to: ● Define anesthetics, anesthesia, local anesthetics & local anesthesia ● Describe the chemistry and classification of local anesthetics ● Describe the differences and or comparison among the local anesthetics ● Describe the uses and mechanism of action of local anesthetics ● State the routes of administration of local anesthetic ● State the doses and effect of overdose of local anesthetics ● Describe the indications, contraindications, complications and management of complications of local anesthetics. ● Describe pharmacological effects, adverse effects, advantages and disadvantages of local anesthesia. ● State the preparation & composition of local anesthetics ● Describe causes of failure to obtain local anesthesia with how to overcome the failure. ● State pharmacology of individual local anesthetic drugs ● Calculate the local anesthetics and vasoconstrictor dosages	1. Local Anesthetics ● Definition of anesthetics, anesthesia, local anesthetics & local anesthesia, chemistry and classification of local anesthetics, comparison among the local anesthetics ● Uses and ,routes of administration of local anesthetic, and doses and effect of overdose of local anesthetics ● Indications, contraindications, complications and management of complications of local anesthetics. ● Pharmacological effects, adverse effects, advantages and disadvantages of local anesthesia, preparation & composition of local anesthetics Causes of failure to obtain local anesthesia with how to overcome the failure. ● Pharmacology of individual local anesthetic drugs, Calculation of local anesthetics and vasoconstrictor dosages	Lecture: 07 hours Tutorial: 03 hours Practical: 01 hours Clinical: 02 hours
● Define sterilization and state the aims and objectives of sterilization ● Classify different methods sterilization ● Describe the different methods of sterilization ● Describe the advantages, disadvantages, differences among the different methods of sterilizations ● Describe the procedure of sterilize dental instruments ● Describe the ideal methods of sterilization in dental clinic ● State the procedure of sterilizing dental hand pieces ● Explain infection control and state the transmissible diseases of concerned to dental surgeons & auxiliaries ● Mention the groups of high risk of contracting hepatitis B ● Describe the methods of infection control in dental clinic	2. Sterilization & Infection Control in Dental Clinic ● Definition of sterilization, aims and objectives of sterilization ● Classification and description of different methods sterilization, advantages, disadvantages, differences among the different methods of sterilizations ● The procedure of sterilization of dental instruments and, the ideal methods of sterilization in dental clinic ● State the procedure of sterilizing dental hand pieces	Lecture: 04 hours Tutorial: 03 hours Practical: 01 hours Clinical: 01 hours

Learning Objectives	Contents	Teaching Hours
Student will be able to: ● Define RCT and state the aims and objectives of RCT ● State the causes of pulp damage and causes of disease of the pulp ● State the systemic aspects of Dental pain ● State the indications & contraindications of RCT ● Describe the RCT of Vital & Non vital tooth ● State the procedure of testing the vitality of tooth ● Explain the single visit & multi-visit RCT ● Describe the steps of RCT ● State the drugs used in RCT for local application & systemic uses ● Describe the properties & functions of drugs used in different steps of RCT ● State the causes of root canal failure ● Describe the precaution for RCT for the patients of infective endocarditis & patient taking steroids	3. Root Canal Therapy (RCT) ● Aims and objectives of RCT ● Causes pulp damage and causes of disease of the pulp ● Systemic aspects of Dental pain ● Indications & contraindications of RCT ● RCT of Vital & Non vital tooth, and procedure of testing the vitality of tooth ● Single visit & multi-visit RCT ● Steps of RCT, and the drugs used in RCT for local application & systemic uses ● Properties & functions of drugs used in different steps of RCT ● State the causes of root canal failure, and precaution for RCT for the patients of infective endocarditis & patient taking steroids	Lecture: 03 hours Tutorial: 02 hours Practical: 01 hours Clinical: 01
● Define & classify antiseptics, disinfectants ● Adverse effect of antiseptics & disinfectants ● State difference between antiseptic & disinfection ● Describe ideal properties of antiseptic & disinfectant ● Mention the uses of antiseptics and disinfectants ● Describe the mechanism of action of antiseptics & disinfectants ● Describe the popularly used antiseptics, disinfectants in dentistry	4. Antiseptic & disinfectants ● Classification of antiseptics, disinfectants ● Adverse effect of antiseptics & disinfectants ● difference between antiseptic & disinfection ● ideal properties of antiseptic & disinfectant ● Uses of antiseptics and disinfectants ● mechanism of action of antiseptics & disinfectants ● popularly used antiseptics, disinfectants in dentistry	Lecture: 04 hours Tutorial: 02 hours Practical: 01 hours

Learning Objectives	Contents	Teaching Hours
Student will be able to: ● Define & classify astringent ,state ideal properties of astringent & styptics ● Mention their dental uses & mode of action ● Describe the popularly used astringents in dentistry ● Mention the dose & administration of astringent & styptics & describe side effect of astringent ● Describe the uses, types, indication & contraindications of gingival retraction cord and other gingival displacement products	5. Astringents and gingival displacements products ● Classification of astringent , ideal properties of astringent & styptics and, their dental uses & mode of action ● The popularly used astringents in dentistry ● Dose & administration of astringent & styptics and side effect of astringent ● Uses, types, indication & contraindications of gingival retraction cord and other gingival displacement products	Lecture: 02 hours Tutorial: 02 hours Practical: 01 hours
● Define & classify mummifying agents ● Describe the popularly used mummifying agents in dentistry ● State the ideal properties, dental uses & mode of action of mummifying agents ● State the uses and adverse effect of mummifying agents ● State the procedure of mummification	6. Mummifying agents ● Classification of mummifying agents and popularly used mummifying agents in dentistry ● Ideal properties, dental uses & mode of action of mummifying agents ● Uses and adverse effect of mummifying agents ● Procedure of mummification	Lecture: 01 hours Tutorial: 01 hours Practical: 01 hours
● Define obtundents ● Classify dental desensitizing agents ● Describe the ideal properties, mode of action, adverse effect and dental uses of desensitizing agents ● Describe the popularly used desensitizing agents in dentistry ● State the mechanism of dentin sensitivity ● State the treatment of dentinal hypersensitivity	7. Dental desensitizing agents and pharmacological control of dentin hypersensitivity ● Definition of obtundents and classification of dental desensitizing agents, the ideal properties, mode of action, adverse effect and dental uses of desensitizing agents ● Popularly used desensitizing agents in dentistry and mechanism of dentin sensitivity ● Treatment of dentinal hypersensitivity	Lecture: 02 hours Tutorial: 02 hours Practical: 01 hours

Learning Objectives	Contents	Teaching Hours
Student will be able to: ● Explain the way of control of dental caries ● Define dental caries ● State the organism & factors responsible for dental caries ● Define and compare different fluoride component ● Describe the uses of fluoride in dentistry ● State the adverse reaction of fluoride ● Describe the mechanism action of fluoride ● State fluoride toxicity ● Perform fluoride application	8. Anti-caries agents and pharmacological control of dental caries ● Way of control of dental caries ● dental caries, organism & factors responsible for dental caries ● compare different fluoride component, uses of fluoride in dentistry, and adverse reaction of fluoride, the mechanism action of fluoride ● fluoride toxicity, fluoride application	Lecture: 03 hours Tutorial: 02 hours Practical: 01 hours
● Define anti-plaque agents ● Describe the mechanism of actions anti-plaque agents ● State the methods of application of anti-plaque agents ● Describe the uses & adverse effects of anti-plaque agents ● Perform application of anti-plaque agents. ● State the organism & factors responsible for periodontal disease ● Explain the way of control of periodontal disease ● Compare different anti-plaque agents ● State the drug treatment of periodontal disease	9. Anti-plaque / anti-gingivitis agents and pharmacological control of periodontal disease ● anti-plaque agents, mechanism of actions anti-plaque agents, methods of application of anti-plaque agents, and uses & adverse effects of anti-plaque agents ● application of anti-plaque agents, control of periodontal disease, and organism & factors responsible for periodontal disease ● Compare different anti-plaque agents, and drug treatment of periodontal disease	Lecture: 03 hours Tutorial: 02 hours Practical: 01 hours Clinical: 01
● Describe the causes of the following dental diseases due to systemic drug treatment: Abnormal haemostasis, Altered host resistant, Angioedema, Coated tongue (black hairy tongue), Dry socket, Dysgeusia, Erythema Multiforme, Gingival enlargement, Leukoplakia & Neutropenia, Lichenoid lesions, Movement disorder, Osteonecrosis Salivary gland enlargement, Sialorrhea, Soft tissue reactions, Xerostomia ● State principles of cancer chemotherapy and state the chemotherapeutic drugs ● State the direct and indirect oral toxic effects of cancer chemotherapy, and complications of cancer radiotherapy	10. Oral manifestations of systemic agents, oral complications of cancer therapy and antineoplastic drugs	Lecture: 03 hours Tutorial: 01 hours

Learning Objectives	Contents	Teaching Hours
Student will be able to: ● Define Haemostatic agents ● Describe the mechanism of actions of Haemostatic agents ● State the methods of application of Haemostatic agents ● Describe the uses and adverse effects of Haemostatic agents ● Perform application of Haemostatic agent ● Describe the uses, types & contraindications of gingival retraction cord ● State the anticuogulants ● Explain the uses of anticuogulant ● Take the necessary action before giving dental treatment of anticuogulant taking patient ● Take precaution before giving surgical dental treatment for a patient taking low dose aspirin	11.Procoagulant, anticoagulant, haemostatics and haemostasis ● Haemostatic agents, mechanism of actions of Haemostatic agents, and methods of application of Haemostatic agents, uses and adverse effects of Haemostatic agents ● Application of Haemostatic agent ● Uses, types & contraindications of gingival retraction cord ● Anticuogulants, uses of anticuogulant and necessary action before giving dental treatment to anticuogulant taking patient ● Precaution before giving surgical dental treatment for a patient taking low dose aspirin	Lecture: 03 hours Tutorial: 02 hours Practical: 01 hours
Student will be able to: ● Define & classify Vitamins ● Describe the source, function, daily requirement, deficiency syndrome and prophylactic uses of different Vitamins ● State the requirements & deficiency syndrome of mineral	12. Vitamins & Minerals ● Classification of Vitamins ● the source, function, daily requirement, deficiency syndrome and prophylactic uses of different Vitamins and, requirements & deficiency syndrome of mineral	Lecture: 02 hours Tutorial: 01 hours
Student will be able to: ● Define Mouth wash & Dentifrices ● State the ideal properties of Mouth wash & Dentifrices ● State the ideal Composition of Mouth wash & Dentifrices ● Classify Mouth wash & Dentifrices ● State the uses and the side effects of Mouth wash & Dentifrices ● Select mouth wash ● State the indication, mechanism of action and side effects of Chlorhexidine containing mouth wash ● State the Doses administration & age limit for mouth wash	13. Mouth Wash & Dentifrices ● Mouth wash & Dentifrices ,the ideal properties of Mouth wash & Dentifrices and ,the ideal Composition of Mouth wash & Dentifrices ● Classification of Mouth wash & Dentifrices, uses and the side effects of Mouth wash & Dentifrices ● indication, mechanism of action and side effects of Chlorhexidine containing mouth wash ● State the Doses administration & age limit for mouth wash	Lecture: 02 hours Tutorial: 02 hours Clinical: 01

Learning Objectives	Contents	Teaching Hours
Student will be able to: ● Define Tooth bleaching ● State the causes of tooth discoloration ● Describe the agents used for tooth bleaching ● State the different methods of tooth bleaching ● State the risk of tooth bleaching	14. Tooth bleaching agents ● Causes of tooth discoloration ● Agents used for tooth bleaching ● Different methods of tooth bleaching and, the risk of tooth bleaching	Lecture: 02 hours Tutorial: 01 hours Practical: 01 hours
● Define & classify Sialogogues & antisialogogues ● State the indications & contraindication of Sialogogues & antisialogogues ● State the causes of Xerostomia ● Describe the Xerogenic drugs ● Describe the causes of increased and decreased salivation ● State the pharmacology of sialogogues and antisialogogues	15. Agents affecting salivation ● Classification of Sialogogues & antisialogogues, indications & contraindication of Sialogogues & antisialogogues ● Causes of Xerostomia, Xerogenic drugs ● Causes of increased and decreased salivation ● Pharmacology of sialogogues and antisialogogues	Lecture: 02 hours Tutorial: 01 hours Practical: xx hours
● State the physiologic, non-physiologic and pharmacologic changes associated with aging ● State the management of fear and anxiety ● Take the necessary measures before giving the dental treatment of the following patients: Anaemia, Leukaemia, Lymphoma, Haemorrhagic disorder, Immunodeficiencies & HIV disease, Liver & kidney disease ● Describe the following dental problems & give the drug treatment of: Dental Hypersensitivity, Acute Pulpitis, Periapical abscess, Periodontal abscess, Cellulitis, Ludwig's angina, Osteomyelitis, Pericoronitis, Gingivitis, Teething, Oral ulcers, Post operative dental pain, Sinusitis, Candidal infection, Thrush, Mercury Dental Amalgam Toxicity	16. Geriatric pharmacology, Treatments for Medically compromised patients and common dental problems ● the physiologic, non-physiologic and pharmacologic changes associated with aging ● the necessary measures before giving the dental treatment of the following patients: Anaemia, Leukaemia, Lymphoma, Haemorrhagic disorder, Immuno-deficiencies & HIV disease, Liver & kidney disease, Pregnant women, Young children, Health and Environment	Lecture: 02 hours Tutorial: 02 hours Practical: 01 hours

Learning Objectives	Contents	Teaching Hours
Student will be able to: ● State the general concepts of pain ● Define Pain, Allodynia, Analgesia, Causalgia, Dyaesthesia, Hyperalgesia, Hyperaesthesia, Hyperpathia, Neuralgia, Trigeminal Neuralgia, Neuritis, Neuropathy, Neuropathic pain, Nociception, Nociceptor, Pain Threshold, Pain tolerance level, Referred pain ● State the Neuroanatomy of pain, Nociceptive pathways, Neurophysiology of pain, Central processing of pain, its control by therapeutic agents, Methods of relieve pain	17. The Pharmacology of Pain and antinociceptive drugs ● General concepts of pain ● Allodynia, Analgesia, Causalgia, Dyaesthesia, Hyperalgesia, Hyperaesthesia, Hyperpathia, Neuralgia, Trigeminal Neuralgia, Neuritis, Neuropathy, Neuropathic pain, Nociception, Nociceptor, Pain Threshold, Pain tolerance level, Referred pain ● Neuroanatomy of pain, Nociceptive pathways, Neurophysiology of pain, Central processing of pain, its control by therapeutic agents, Methods of relieve pain	Lecture: 02 hours Tutorial: 02 hours Practical: xx hours
● State the emergency prevention, emergency preparedness, emergency drugs ● Manage medical emergencies of relevance to dental practice of the following: ○ Syncope ○ Angina Pectoris, Myocardial infarction, Cardiac arrest ○ Hypertension, hypotension ○ Hemorrhage ○ Cerebrovascular accident ○ Asthma ○ Insulin shock / diabetic coma ○ Anaphylaxis / other acute allergic reactions	18. Drugs for Medical Emergencies in Dental Practice and Management ● Emergency prevention, emergency preparedness, emergency drugs ● Medical emergencies of relevance to dental practice of the following: ○ Syncope ○ Angina Pectoris, myocardial infarction, Cardiac arrest ○ Hypertension, hypotension ○ Hemorrhage ○ Cerebrovascular accident ○ Asthma ○ Insulin shock / diabetic coma ○ Anaphylaxis / other acute allergic reactions	Lecture: 03 hours Tutorial: 02 hours Practical: 02 hours

Professional Examination:

Marks distribution of Assessment of General Pharmacology (Group A) (Total marks 100)

- Written = 50 (SAQ =35 + Formative assessment = 5+ MCQ = 10)
- SOE = 50
- Practical = 50

PRACTICAL MARKS DISTRIBUTION

Total Marks – 50 (general pharmacology)

A. OSPE:

Marks-20

01. Question Stations: Marks (5x4) =20

- | | | |
|-------------|---|--|
| STATION- 01 | } | – Sources of Drug, Drug Formulations (Dosage forms) & Drug |
| STATION- 02 | | – Delivery System |
| STATION- 03 | | – Identification of Drug |
| STATION- 04 | | – Selection of P drugs/Essential drug Concept / |
| STATION -05 | | – Principle of Rational Prescribing / Drug information sources etc |

B.TRADITIONAL

Marks - 30

1. Prescription writing - 01 [Format of Ideal prescription]

Marks - 10

2. Drug Interactions - 01

Marks - 10

3. Practical note book

Marks- 10

Marks distribution of Assessment of Dental Therapeutics (Group B) (Total marks 100)

- Written = 50 (SAQ =35 + marks of formative assessment = 5+ MCQ = 10)
- SOE = 50
- Practical = 50

PRACTICAL MARKS DISTRIBUTION

Total Marks – 50 Dental Pharmacology)

A. OSPE: Marks-20

01. Question Stations:Marks (5x4) =20

STATION- 01	}	Identification of Drugs, Sources of the drugs, Indications, Contraindications, Uses, Properties, Mode of action
STATION-02		
STATION-03		
STATION- 04		
STATION -05		

B.TRADITIONAL

Marks - 30

1. Prescription writing - 01 [Format of Dental Prescription]

Marks - 10

2. Hands on practical work- 01

Marks - 10

3. Practical note book

Marks- 10